

ASSIGNMENT #8 10/28/22

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SECTION 5.1

Exercise (5.1.4 — Stefano Fochesatto). Let A and B be finite groups and let p be prime. Prove that any Sylow p -subgroup of $A \times B$ is of the form $P \times Q$, where $P \in \text{Syl}_p(A)$ and $Q \in \text{Syl}_p(B)$. Prove that $n_p(A \times B) = n_p(A)n_p(B)$. Generalize both of these results to a direct product of any finite number of finite groups.

Proof. Suppose A and B are finite groups and let p be prime. We know that $|A| = p^\alpha m$ and $|B| = p^\beta n$ where $p \nmid m, n$. Note that by properties of direct products we know that $|A \times B| = p^{\alpha+\beta} mn$. Consider $P \in \text{Syl}_p(A)$ and $Q \in \text{Syl}_p(B)$ and note that $|P \times Q| = p^{\alpha+\beta} mn$ and $p \nmid mn$, therefore $P \times Q \in \text{Syl}_p(A \times B)$. By Sylow's Theorem we know every $H \in \text{Syl}_p(A \times B)$ is conjugate to $P \times Q$. Let $g \in A$ and $h \in B$ then there exists some $(g, h) \in (A \times B)$ such that $H = (g, h)(P, Q)(g, h)^{-1} = (gPg^{-1}, hQh^{-1})$. By Sylow's Theorem, we also know that $gPg^{-1} \in \text{Syl}_p(A)$ and similarly $hQh^{-1} \in \text{Syl}_p(B)$. Therefore every $H \in \text{Syl}_p(A \times B)$ is of the form $(P \times Q)$ where $P \in \text{Syl}_p(A)$ and $Q \in \text{Syl}_p(B)$.

Since every $H \in \text{Syl}_p(A \times B)$ is of the form $(P \times Q)$ where $P \in \text{Syl}_p(A)$ and $Q \in \text{Syl}_p(B)$ we know that there are only $n_p(A)n_p(B)$ ways to combine Sylow p -subgroups to get a $(P \times Q)$ product. Thus $n_p(A \times B) = n_p(A)n_p(B)$.

Let $G_1, G_2, \dots, G_n$ be groups and let $G = G_1 \times G_2 \times \dots \times G_n$, then any $X \in \text{Syl}_p(G)$ is of the form $X = P_1 \times P_2 \times \dots \times P_n$ where $P_i \in \text{Syl}_p(G_i)$ and $n_p(G) = \prod n_p(G_i)$.

□

Exercise (5.1.5 — Glen Woodworth). Exhibit a nonnormal subgroup of $Q_8 \times Z_4$ (note that every subgroup of each factor is normal).

Proof. Consider the subgroup $\langle(-k, 0)\rangle = \{(-k, 0), (1, 0)\}$ in $Q_8 \times Z_4$. Observe,

$$\begin{aligned} (j, 0)(-k, 0)(-j, 0) &= (j(-k), 0)(-j, 0) = (-jk, 0)(-j, 0) = (i, 0)(-j, 0) \\ &= (i(-j), 0) = (-ij, 0) = (k, 0) \notin \langle(-k, 0)\rangle. \end{aligned}$$

Hence, $\langle(-k, 0)\rangle$ is not a normal subgroup of $Q_8 \times Z_4$. □

Exercise (5.1.10 — Noah Palmer). Let p be a prime. Let A and B be two cyclic groups of order p with generators x and y , respectively. Set $E = A \times B$ so that E is the elementary abelian group of order $p^2 : E_{p^2}$. Prove that the distinct subgroups of E of order p are

$$\langle x \rangle, \quad \langle xy \rangle, \quad \dots, \quad \langle xy^{p-1} \rangle, \quad \langle y \rangle$$

Proof. Let p be a prime, $\langle x \rangle = A$ and $\langle y \rangle = B$ be two cyclic groups of order p , and set $E = A \times B$ so that E is the elementary abelian group of order p^2 . Since A and B are cyclic groups of prime order it follows that each element of A and B have order p . Hence, each of the subgroups

$$\langle x \rangle, \quad \langle xy \rangle, \quad \dots, \quad \langle xy^{p-1} \rangle, \quad \langle y \rangle$$

will have order p . Additionally, each of these groups are unique since each y^i is unique within its own cyclic group which implies that the subgroups generated by $\langle(1, 1, \dots, x, \dots, y^i, \dots 1)\rangle$ will be unique for each $1 \leq i < p$. From example 3 in section 5.1 of Dummit and Foote the number of subgroups of order p in an elementary abelian group of order p^2 is $p + 1$. Since, we have given a list of exactly $p + 1$ distinct groups, we're done. □

SECTION 5.2

Exercise (5.2.2 — Oleksandr Bobrovnikov). In each of parts (a) to (d) give the lists of invariant factors for all abelian groups of the specified order:

(a) order 270:

Solution: Note that $270 = 2 \times 3^3 \times 5$. Since $2 \times 3 \times 5 \mid n_1$, then $n_1 \in \{2 \times 3 \times 5, 2 \times 3^2 \times 5, 2 \times 3^3 \times 5\}$. If $n_1 = 2 \times 3 \times 5$ then from $n_{i+1} \mid n_i$ we get $n_2 = 3$ and $n_3 = 3$. If $n_1 = 2 \times 3^2 \times 5$ then similarly we get $n_2 = 3$. If $n_1 = 2 \times 3^3 \times 5$ then this is the only invariant factor. $\square$

(b) order 9801:

Solution: Note that $9801 = 3^4 \times 11^2$. Thus

$$n_1 \in \{3 \times 11, 3^2 \times 11, 3^3 \times 11, 3^4 \times 11, 3 \times 11^2, 3^2 \times 11^2, 3^3 \times 11^2, 3^4 \times 11^2\}.$$

If $n_1 = 3 \times 11$ then $n_2 = 3 \times 11$ since otherwise we will not obtain $n = \prod n_j$ (if we take 3 for example, there will be not enough 11). Then list of invariant factors is $3 \times 11, 3 \times 11, 3, 3$.

If $n_1 = 3^2 \times 11$ then similarly n_2 should be 3×11 or $3^2 \times 11$. Then possible lists are the following: $3^2 \times 11, 3 \times 11, 3$ and $3^2 \times 11, 3^2 \times 11$.

If $n_1 = 3^3 \times 11$ then the only possible list is $3^3 \times 11, 3 \times 11$

If $n_1 = 3^4 \times 11$ then $n_2 = 11$.

If $n_1 = 3 \times 11^2$ then the only possible list is $n_1, 3, 3, 3$.

If $n_1 = 3^2 \times 11^2$ then possible lists are $n_1, 3, 3$ and $n_1, 3^2$.

If $n_1 = 3^4 \times 11^2$ then the list is $n_1, 3$.

And finally if $n_1 = 9801$, then this is the entire list. $\square$

(c) order 320:

Solution: Note that $320 = 2^6 \times 5$. So $n_1 = 2^j \times 5$, where $1 \leq j \leq 6$. So possible list are: $(2 \times 5, 2, 2, 2, 2, 2)$, $(2^2 \times 5, 2^2, 2^2)$, $(2^2 \times 5, 2^2, 2, 2)$, $(2^2 \times 5, 2, 2, 2, 2)$, $(2^3 \times 5, 2^3)$, $(2^3 \times 5, 2^2, 2)$, $(2^3 \times 5, 2, 2, 2)$, $(2^4 \times 5, 2^2)$, $(2^4 \times 5, 2, 2)$, $(2^5 \times 5, 5)$, (2^6) $\square$

(d) order 105:

Solution: Note that $105 = 3 \times 5 \times 7$. Then the only possible list is $n_1 = 105$. $\square$

Exercise (5.2.4 — Noah Palmer). In each of parts (a) to (d) determine which pairs of abelian groups listed are isomorphic.

(a) $\{4, 9\}$, $\{6, 6\}$, $\{8, 3\}$, $\{9, 4\}$, $\{6, 4\}$, $\{64\}$.

- (b) $\{2^2, 2 \cdot 3^2\}, \{2^2 \cdot 3, 2 \cdot 3\}, \{2^3 \cdot 3^2\}, \{2^2 \cdot 3^2, 2\}.$
- (c) $\{5^2 \cdot 7^2, 3^2 \cdot 5 \cdot 7\}, \{3^2 \cdot 5^2 \cdot 7, 5 \cdot 7^2\}, \{3 \cdot 5^2, 7^2, 3 \cdot 5 \cdot 7\}, \{5^2 \cdot 7, 3^2 \cdot 5, 7^2\}.$
- (d) $\{2^2 \cdot 5 \cdot 7, 2^3 \cdot 5^3, 2 \cdot 5^2\}, \{2^3 \cdot 5^3 \cdot 7, 2^3 \cdot 5^3\}, \{2^2, 2 \cdot 7, 2^3, 5^3, 5^3\}, \{2 \cdot 5^3, 2^2 \cdot 5^3, 2^3, 7\}.$

Proof. We use the fact that $Z_m \times Z_n \cong Z_{mn}$ if and only if $\gcd(m, n) = 1$ to rearrange the abelian groups into isomorphic pairs, where possible.

- (a) Note that $Z_4 \times Z_9 \cong Z_9 \times Z_4$. Thus, $\{4, 9\} \cong \{9, 4\}.$
- (b) Note that $Z_{2^2} \times Z_{2 \cdot 3^2} \cong Z_{2^2 \cdot 3^2} \times Z_2$. Thus, $\{2^2, 2 \cdot 3^2\} \cong \{2^2 \cdot 3^2, 2\}.$
- (c) Note that $Z_{5^2 \cdot 7^2} \times Z_{3^2 \cdot 5 \cdot 7} \cong Z_{3^2 \cdot 5^2 \cdot 7} \times Z_{5 \cdot 7^2}$. Also, $Z_{3^2 \cdot 5^2 \cdot 7} \times Z_{5 \cdot 7^2} \cong Z_{5^2 \cdot 7} \times Z_{3^2 \cdot 5} \times Z_{7^2}$. Thus, $\{5^2 \cdot 7^2, 3^2 \cdot 5 \cdot 7\} \cong \{3^2 \cdot 5^2 \cdot 7, 5 \cdot 7^2\} \cong \{5^2 \cdot 7, 3^2 \cdot 5, 7^2\}.$
- (d) Note that $Z_{2^2} \times Z_{2 \cdot 7} \times Z_{2^3} \times Z_{5^3} \times Z_{5^3} \cong Z_{2^2 \cdot 5^3} \times Z_{2 \cdot 7 \cdot 5^3} \times Z_{2^3} \cong Z_{2 \cdot 5^3} \times Z_{2^2 \cdot 5^3} \times Z_{2^3} \times Z_7$. Therefore, $\{2^2, 2 \cdot 7, 2^3, 5^3, 5^3\} \cong \{2 \cdot 5^3, 2^2 \cdot 5^3, 2^3, 7\}.$

□

Exercise (5.2.9 — Noah Palmer). Let $A = Z_{60} \times Z_{45} \times Z_{12} \times Z_{36}$. Find the number of elements of order 2 and the number of subgroups of index 2 in A .

Proof. Begin by rewriting the direct product in its invariant form $Z_{60} \times Z_{45} \times Z_{12} \times Z_{36} \cong Z_{180} \times Z_{60} \times Z_3 \times Z_{36} \cong Z_{180} \times Z_{180} \times Z_{12} \times Z_3$. Now, note that each of Z_{180} and Z_{12} has an element of order 2 whereas Z_3 does not. Hence, there will be 2^3 elements of order less than or equal to 2. From these we must exclude the identity since it has order 1. Thus, there are 7 elements of order 2.

To find the subgroups of index 2 in A consider the elementary divisor factorization of $A = Z_5 \times Z_5 \times Z_4 \times Z_4 \times Z_4 \times Z_9 \times Z_9 \times Z_3 \times Z_3$. Now, any subgroup of order 583,200 will have index 2 in A . From the above factorization, and the prime factorization of 583,200, these subgroups will have the form $A' \times Z_5 \times Z_5 \times Z_9 \times Z_9 \times Z_3 \times Z_3$ where A' is a subgroup of $Z_4 \times Z_4 \times Z_4$ with order 32. There are three ways to select such a subgroup (e.g. $Z_4 \times Z_4 \times Z_2$, $Z_2 \times Z_4 \times Z_4$, and $Z_4 \times Z_2 \times Z_4$). Thus, there are three subgroups of index 2 in A .

□

SECTION 5.4

Exercise (5.4.1 — Stefano Fochesatto). Prove that if $x, y \in G$ then $[y, x] = [x, y]^{-1}$. Deduce that for any subsets A, B of G , $[A, B] = [B, A]$

Proof. Suppose if $x, y \in G$ then we know $[y, x] = y^{-1}x^{-1}yx$. Note that $y^{-1}x^{-1}yx = ((xy)^{-1})(yx) = ((yx)^{-1}(xy))^{-1} = (x^{-1}y^{-1}xy)^{-1} = [x, y]^{-1}$.

Let $A, B \leq G$ and consider the group generated by $[A, B]$. By definition we know that $[A, B] = \langle [a, b] | a \in A, b \in B \rangle$. Since $[A, B]$ is a group generated by a set i.e the set of all commutators, it must also be generated by a set of commutator inverses, $[A, B] = \langle [a, b]^{-1} | a \in A, b \in B \rangle$ By the previous result we know that $[a, b]^{-1} = [b, a]$ so $[A, B] = \langle [b, a] | a \in A, b \in B \rangle = [B, A]$. □

Exercise (5.4.11 — Oleksandr Bobrovnikov). Prove that if $G = HK$ where H and K are characteristic subgroups of G with $H \cap K = 1$ then $\text{Aut}(G) \cong \text{Aut}(H) \times \text{Aut}(K)$. Deduce that if G is an abelian group of finite order then $\text{Aut}(G)$ is isomorphic to the direct product of the automorphism groups of its Sylow subgroups.

Solution: Since H and K are normal subgroups of G with $H \cap K = 1$, we know that $HK \cong H \times K$. Let $G = H \times K$. We define $\varphi_H \in \text{Aut}(H)$ by $\varphi_H(h) = \pi_H(\varphi(h, 1))$ and similarly we define $\varphi_K \in \text{Aut}(K)$. Then consider $\Phi : \text{Aut}(G) \rightarrow \text{Aut}(H) \times \text{Aut}(K)$, which acts by $\varphi \mapsto (\varphi_H, \varphi_K)$. This is an isomorphism. First note that it is a homomorphism:

$$(1) \quad [\Psi(\varphi \circ \psi)](h, k) =$$

$$((\varphi \circ \psi)_H(h), (\varphi \circ \psi)_K(k)) = (\pi_H[(\varphi \circ \psi)(h, 1)], \pi_K[(\varphi \circ \psi)(1, k)]) =$$

$$((\pi_H \circ \varphi \circ \pi_H \circ \psi)(h, 1), (\pi_K \circ \varphi \circ \pi_K \circ \psi)(1, k)) = ((\varphi_H \circ \psi_H)(h, 1), (\varphi_K \circ \psi_K)(1, k)) =$$

$$[\Psi(\varphi) \circ \Psi(\psi)](h, k),$$

and similarly $\Psi(\varphi^{-1}) = \Psi(\varphi)^{-1}$. Finally, Ψ is a bijection since by φ_H and φ_K one can uniquely determine φ .

If G is an abelian group of finite order then every its Sylow p -subgroup are normal and thus are characteristic, and intersection of any two of them is identity. Then using induction by number of these subgroups one may show that $\text{Aut}(G)$ is isomorphic to the direct product of the automorphism groups of its Sylow subgroups. Base is proven above, and a step uses base and associativity of direct product.

□

Exercise (5.4.17 — Glen Woodworth). If K is normal subgroup of G and K is cyclic, prove that $G' \leq C_G(K)$. [Recall that the automorphism group of a cyclic group is abelian.]

Proof. Let $K \trianglelefteq G$. Recall that the automorphism group of a cyclic group is abelian. Since K is cyclic, $\text{Aut}(K)$ is abelian. Consider the map

$$\Phi : G \rightarrow \text{Aut}(K)$$

$$g \mapsto \phi_g : k \mapsto gkg^{-1}.$$

Since $K \trianglelefteq G$, $gkg^{-1} \in K$ for all $k \in K$ and for all $g \in G$. Thus, Φ is well-defined. We now show that Φ is a homomorphism. Let $g_1, g_2 \in G$ and $k \in K$. Observe,

$$\begin{aligned} \Phi(g_1g_2)(k) &= \phi_{g_1g_2}(k) = g_1g_2k(g_1g_2)^{-1} = g_1g_2kg_2^{-1}g_1^{-1} = \phi_{g_1}(g_2kg_2^{-1}) \\ &= \phi_{g_1}(\phi_{g_2}(k)) = (\phi_{g_1} \circ \phi_{g_2})(k) = (\Phi(g_1) \circ \Phi(g_2))(k). \end{aligned}$$

Thus, Φ is a homomorphism. We now show that $\ker \Phi = C_G(K)$. Let $x \in C_G(K)$. Then

$$\Phi(x)(k) = xkx^{-1} = k.$$

Thus, $\Phi(x) = \text{Id}$. Suppose $y \in \ker \Phi$. We have

$$k = \Phi(y)(k) = yky^{-1}.$$

Therefore, $y \in C_G(K)$. Hence, $\ker \Phi = C_G(K)$. Proposition 7 from Section 5.4 states that if $\phi : G \rightarrow A$ is any homomorphism where G is a group and A is an abelian group, then $G' \leq \ker \phi$. Since $\text{Aut}(K)$ is abelian, and Φ is a homomorphism with $\ker \Phi = C_G(K)$, we have $G' \leq \ker \Phi = C_G(K)$. $\square$

SECTION 5.5

Exercise (5.5.1 — Oleksandr Bobrovnikov). Prove that $C_K(H) = \ker \varphi$ (recall that $C_K(H) = C_G(H) \cap K$).

Proof. Note that $\ker \varphi = \{(1, k) \in G \mid \forall h \in H : k \cdot h = h\}$ and $C_K(H) = C_G(H) \cap K = \{(1, k) \in G \mid \forall (h, s) \in G : (h, s)(1, k) = (1, k)(h, s)\}$. The second is equivalent to $C_K(H) = \{(1, k) \in G \mid \forall (h, s) \in G : (hs \cdot 1, sk) = (1k \cdot h, ks)\}$. From this follows that $C_K(H) = \ker \varphi$. $\square$

Exercise (5.5.2 — Stefano Fochesatto). Let H and K be groups, let φ be a homomorphism from K into $\text{Aut}(H)$ and identify H and K as subgroups of $G = H \rtimes_{\varphi} K$. Prove that $C_H(K) = N_H(K)$

Proof. By definition we know that $C_H(K) \subseteq N_H(K)$. Let $(h, 1) \in N_H(K)$. By definition we know that for all $(1, k_1) \in K$ there exists some $(1, k_2) \in K$ such that $(h, 1)(1, k_1)(h^{-1}, 1) = (1, k_2)$. By multiplication in G we get $(h, 1)(\varphi_{k_1}(h^{-1}), k_1) = (1, k_2)$. Again multiplication in G gives us, $(\varphi_{k_1}(h)\varphi_{k_1}(h^{-1}), k_1) = (1, k_2)$. So $k_1 = k_2$ and by we can get $(h, 1) \in C_H(K)$. Thus $C_H(K) = N_H(K)$. $\square$

Exercise (5.5.8 — Glen Woodworth). Construct a non-abelian group of order 75. Classify all groups of order 75 (there are three of them). [Use Exercise 6 to show that the non-abelian group is unique.] (The classification of groups of order pq^2 , where p and q are primes with $p < q$ and p not dividing $q - 1$, is quite similar.)

Proof. Consider the homomorphism

$$\begin{aligned}\Phi : Z_{25} &\rightarrow \text{Aut}(Z_3) \\ \bar{0} &\mapsto \text{Id} : \bar{1} \mapsto \bar{1} \\ \bar{1} &\mapsto \phi : \bar{1} \mapsto \bar{2}.\end{aligned}$$

We show that Φ is well-defined. Observe,

$$\phi(\bar{1} + \bar{2}) = \phi(\bar{0}) = \bar{0} = \bar{2} + \bar{1} = \phi(\bar{1}) + \phi(\bar{2})$$

Thus, ϕ is a homomorphism. Since ϕ is obviously a bijection, $\phi \in \text{Aut}(Z_3)$. Thus, Φ is well-defined. We now show Φ is a homomorphism. First, observe that since $\Phi(\bar{1})(\bar{1}) = \phi(\bar{1}) = \bar{2}$ and since there are only two options in the image, if $x \in Z_{25}$ is odd, then $\Phi(x)(\bar{1}) = \phi(\bar{1}) = \bar{2}$. Similarly, if $y \in Z_{25}$ is even, then $\Phi(y)(\bar{1}) = \text{Id}(\bar{1}) = \bar{1}$. Now, suppose $g_1, g_2 \in Z_{25}$. We have three cases to consider. For the first case, suppose g_1 and g_2 are both even. Then $g_1 + g_2$ is even. Observe,

$$\Phi(g_1 + g_2)(\bar{1}) = \text{Id}(\bar{1}) = \text{Id}(\text{Id}(\bar{1})) = \Phi(g_1)(\Phi(g_2)(\bar{1})) = (\Phi(g_1) \circ \Phi(g_2))(\bar{1}).$$

For the second case, suppose g_1 and g_2 are both odd. Then $g_1 + g_2$ is even. We have

$$\Phi(g_1 + g_2)(\bar{1}) = \text{Id}(\bar{1}) = \bar{1} = \phi(\bar{2}) = \phi(\phi(\bar{1})) = \Phi(g_1)(\Phi(g_2)(\bar{1})) = (\Phi(g_1) \circ \Phi(g_2))(\bar{1}).$$

For the last case, suppose g_1 is even and g_2 is odd. Then $g_1 + g_2$ is odd. Observe,

$$\Phi(g_1 + g_2)(\bar{1}) = \phi(\bar{1}) = \text{Id}(\phi(\bar{1})) = \Phi(g_1)(\Phi(g_2)(\bar{1})) = (\Phi(g_1) \circ \Phi(g_2))(\bar{1}).$$

Thus, Φ is a well-defined homomorphism. Hence, by the definition of semidirect product and Theorem 10, $G = Z_3 \rtimes_{\Phi} Z_{25}$ is a group of order 75. Let $\cdot$ denote the left action of Z_{25}

on Z_3 determined by Φ . To see that G is nonabelian, consider $(\bar{1}, \bar{3}), (\bar{2}, \bar{4}) \in G$. Observe,

$$\begin{aligned}
 (\bar{1}, \bar{3})(\bar{2}, \bar{4})(\bar{1}, \bar{3})^{-1} &= (\bar{1} + \bar{3} \cdot \bar{2}, \bar{3} + \bar{4})(\bar{2}\bar{2} \cdot \bar{2}, \bar{2}\bar{2}) = (\bar{1} + \Phi(\bar{3})(\bar{2}), \bar{7})(\Phi(\bar{2}\bar{2})(\bar{2}), \bar{2}\bar{2}) \\
 &= (\bar{1} + \phi(\bar{2}), \bar{7})(\text{Id}(\bar{2}), \bar{2}\bar{2}) = (\bar{1} + \bar{1}, \bar{7})(\bar{2}, \bar{2}\bar{2}) = (\bar{2}, \bar{7})(\bar{2}, \bar{2}\bar{2}) \\
 &= (\bar{2} + \bar{7} \cdot \bar{2}, \bar{7} + \bar{2}\bar{2}) = (\bar{2} + \Phi(\bar{7})(\bar{2}), \bar{4}) = (\bar{2} + \phi(\bar{2}), \bar{4}) = (\bar{2} + \bar{1}, \bar{4}) \\
 &= (\bar{0}, \bar{4}).
 \end{aligned}$$

Since $(\bar{2}, \bar{4}) \neq (\bar{0}, \bar{4})$, G is nonabelian. We are given that there are only three groups of order 75. The other two groups of order 75 are Z_{75} and $Z_{15} \times Z_5$, which are both abelian. Note that since $(5, 15) \neq 1$, Z_{75} is not isomorphic to $Z_{15} \times Z_5$. Exercise 6 of section 5.5 states that if there exists another homomorphism Φ_1 from Z_{25} to $\text{Aut}(Z_3)$ such that $\Phi(Z_{25})$ and $\Phi_1(Z_{25})$ are conjugate subgroups of $\text{Aut}(Z_3)$, then $Z_3 \rtimes_{\Phi} Z_{25} \cong Z_3 \rtimes_{\Phi_1} Z_{25}$. If we suppose that Φ_1 is non-trivial, then $\phi \in \Phi_1(Z_{25})$ because ϕ is the only non-identity element in $\text{Aut}(Z_3)$. Since $\text{Id} \in \Phi_1(Z_{25})$ by the definition of a homomorphism, $\Phi_1(Z_{25}) = \Phi(Z_{25})$. By Exercise 6, $Z_3 \rtimes_{\Phi} Z_{25} \cong Z_3 \rtimes_{\Phi_1} Z_{25}$. $\square$